MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SG7 AND 12SG7 1/

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: **Pentode, RF semiremote cutoff**

Outline --- 8-1 (EIA)

Base --- B8-21 (phenolic)

Envelope --- MT-8

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	sh int sd	h	k	g1	k	g2	h	a

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec2	Ec1	Pp	Pg2	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	W	W	v	ft
Maximum:								
Type 6SG7	6.9	330	220	---	3.3	0.7	100	(See note 1)
Type 12SG7	13.9	330	220	---	3.3	0.7	100	(See note 1)
Minimum:								
Type 6SG7	5.7	---	---	---	---	---	---	---
Type 12SG7	11.3	----	---	---	---	---	---	---

TEST CONDITIONS:

Type 6SG7	6.3	250	125	-1.0	---	---	---	---
Type 12SG7	12.6	250	125	-1.0	---	---	---	---

GENERAL:

Qualification - Required

1/ **Formerly Tube Types 6SG7Y and 12SG7Y**

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Ec1 = -1.5 Vdc See note 2	0.65	II	Ic1	0	-1.0	μAdc
1256	Electrode current (anode)		0.65	II	Ib	9.0	14.6	mAdc
1306	Transconductance (1)		0.65	II	Sm	3,800	5,600	μmhos
1231	Emission	Eb = Ec2 = Ec1 = 20 Vdc, See note 2	0.65	II	Is	95	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	---	---	Ep	---	400	mVac
1301	Heater current Type 12S97		---	---	If	275	325	mA
1301	Heater current Type 12S97		---	---	If	138	162	mA
1336	Heater-cathode leakage		---	---	Ihk	---	20	μAdc
1261	Electrode current (screen)		---	---	Ic2	2.9	5.9	mAdc
1306	Transconductance (2)	Ec1 = -14 Vdc	---	---	Sm	10	100	μmhos
1331	Direct-interelectrode capacitance		---	---	Cgp Cin Cout	--- 6.8 4.9	0.004 10.2 9.1	pF pF pF
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1111	Base pin solder depth		---	---	---	---	---	---
1211	Insulation of electrodes		4.0	S3	---	---	---	---
1151	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
----	Life-test provisions	Group A; Ehk = 100 V	---	---	---	---	---	---
----	Life-test end point (500 hours)							
1306	Transconductance (1)		---	---	Sm	3,050	---	μmhos

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.

Custodians:

Army - EL
Navy - EC
Air Force - 80

Review activities:

Army - EL
Navy -
Air Force - 17
DSA - ES

User activities:

Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:

Air Force - 80

(Project 5960-2708)

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OMB Approval
No. 22-R255

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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